Project: High Level Architecture Design - Data Flow Diagram & Component Level Design - Sequence or Workflow

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1. High Level Architecture Design

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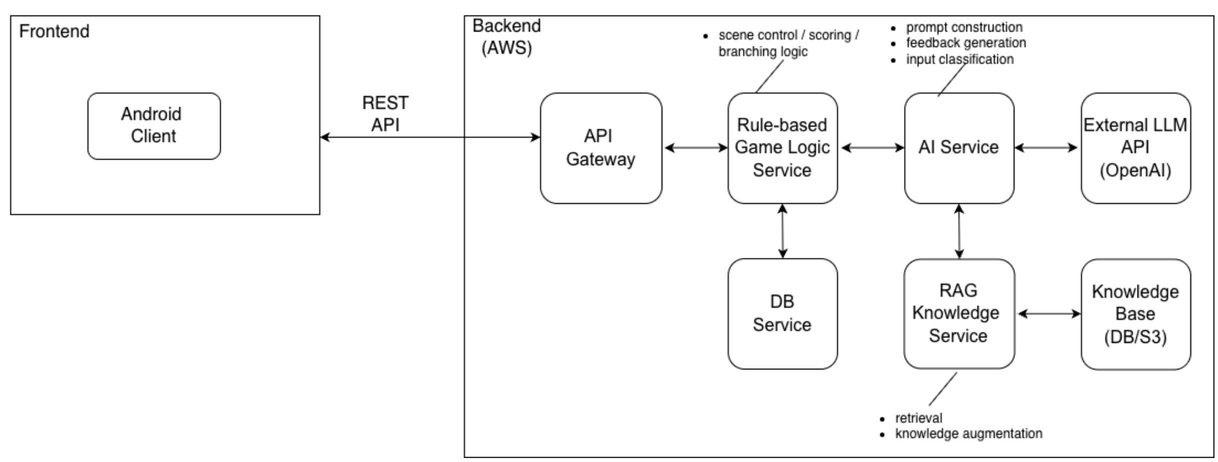


The high-level architecture of the project is designed as a hybrid cloud-based system consisting of an Android frontend client and an AWS-hosted backend. The frontend is responsible for presenting story scenes, collecting player choices or free-text input, and displaying AI-generated feedback and knowledge query results. Communication between the Android client and the backend is handled through REST APIs.

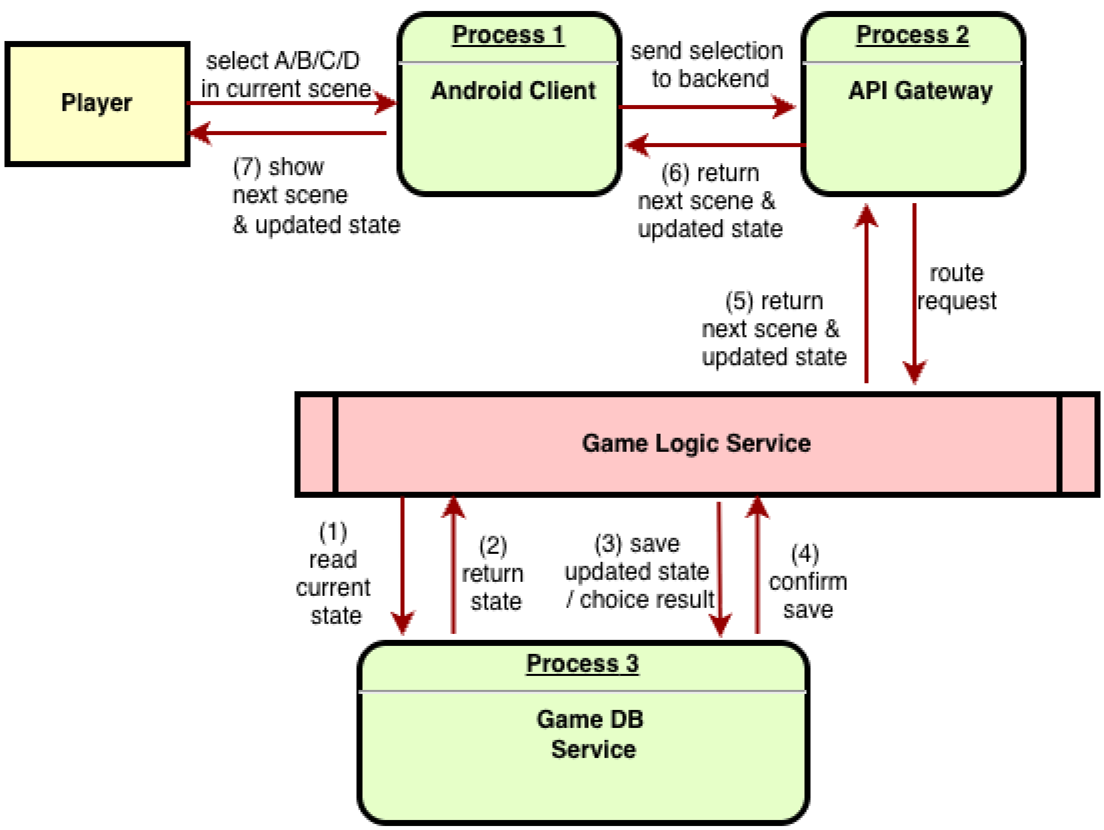
On the backend, the system is divided into several major components. The API Gateway serves as the entry point for all client requests. The Rule-based Game Logic Service manages deterministic gameplay functions such as scene progression, branching logic, score calculation, and ending evaluation. This ensures that the core game rules remain stable and controllable.

To enhance interactivity and immersion, the architecture also includes an AI Service, which is responsible for prompt construction, narrative feedback generation, and free-text input classification. This service communicates with an external LLM API (OpenAI) to produce context-aware responses. In addition, a RAG Knowledge Service is used to support the in-game knowledge assistant. It retrieves relevant information from the Knowledge Base and augments the context before generating responses. The DB Service stores structured game-related data such as player progress, scene states, and system records, while the Knowledge Base (DB/S3) stores lore documents and retrieval resources for the RAG pipeline.

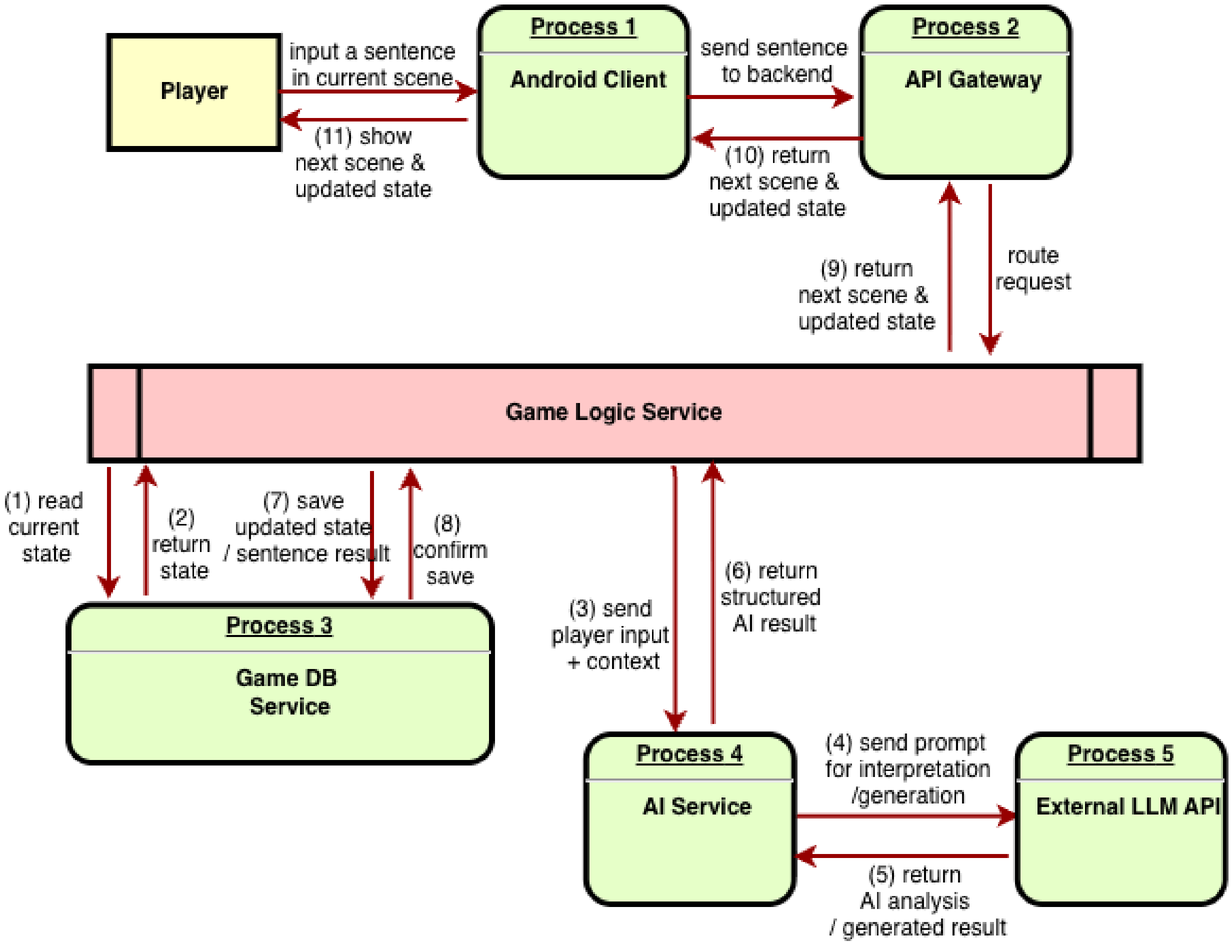
Note: Authentication and user management are considered out of scope for the current prototype but can be integrated using services such as AWS Cognito in future iterations.

2. Data Flow Diagram & Component Level Design

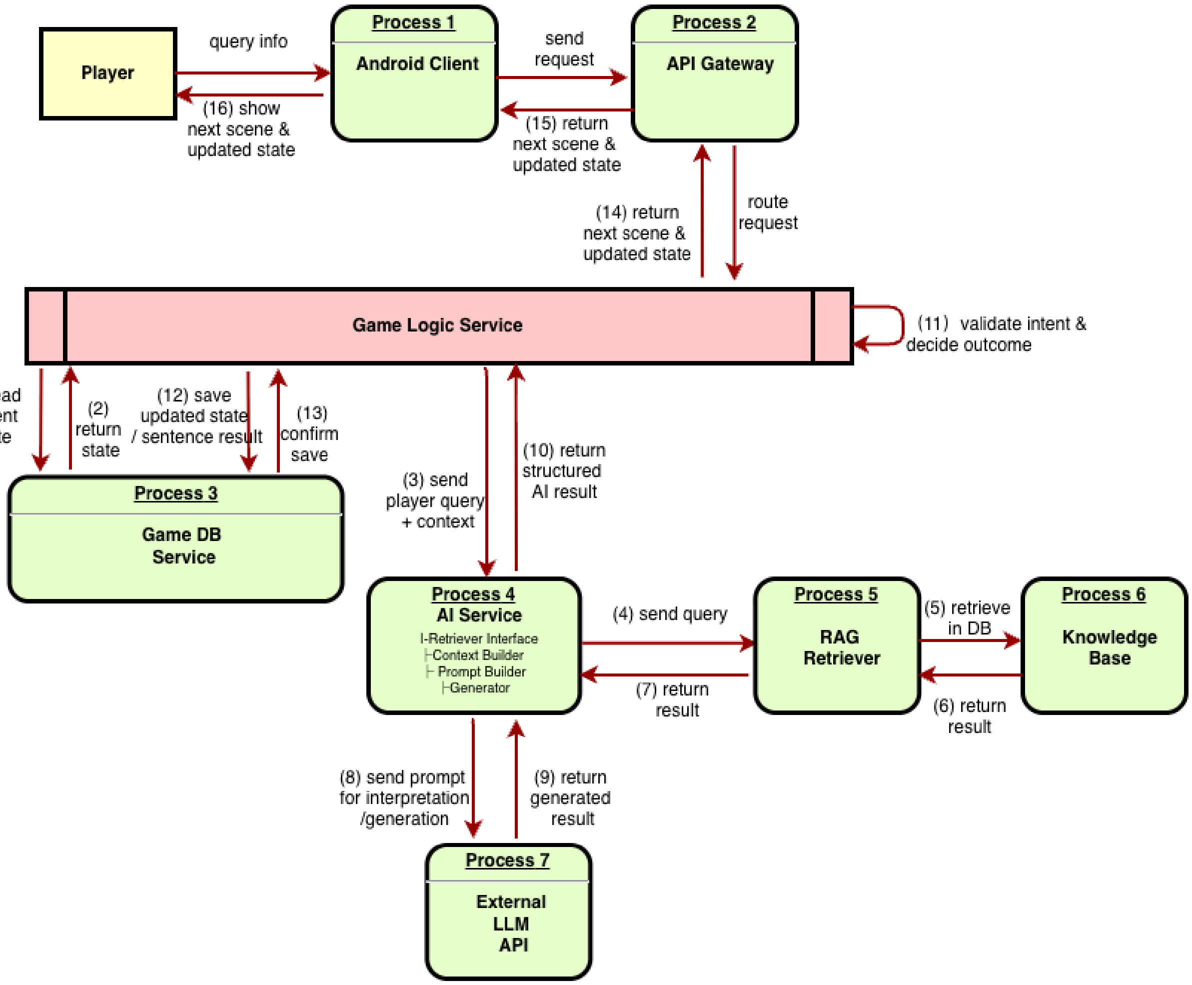
Scene 1: User selects A/B/C/D option.



Scene 2: User inputs free_text in current scene.



Scene 3: User query information from Hermione's notes.



3. Time Sequence Diagram